

# Exponentials & Polynomials

## ANSWER KEY



### Laws of exponents

- 1 Simplify  $x^5 \cdot x^3$ .  
 $x^5 \cdot x^3 = x^{5+3} = x^8$ .
  - 2 Simplify  $\frac{y^9}{y^4}$ .  
 $\frac{y^9}{y^4} = y^{9-4} = y^5$ .
  - 3 Simplify  $(2x^3)^4$ .  
 $(2x^3)^4 = 2^4 x^{12} = 16x^{12}$ .
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### Negative and fractional exponents

- 4 Simplify  $x^{-3}$ , writing your answer without a negative exponent.  
 $x^{-3} = \frac{1}{x^3}$ .
  - 5 Simplify  $16^{3/4}$ .  
 $16^{3/4} = (16^{1/4})^3 = 2^3 = 8$  (since  $16^{1/4} = 2$ , because  $2^4 = 16$ ).
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### Adding, subtracting, and multiplying polynomials

- 6 Simplify  $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$ .  
Combine like terms:  $(3x^2 + x^2) + (2x - 4x) + (-5 + 7) = 4x^2 - 2x + 2$ .
  - 7 Expand  $(x + 3)(x - 5)$ .  
 $(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15$ .
  - 8 Expand  $(2x - 1)^2$ .  
 $(2x - 1)^2 = (2x)^2 - 2(2x)(1) + 1^2 = 4x^2 - 4x + 1$ .
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### Factoring polynomials

- 9 Factor  $x^2 - 16$  completely.  
This is a difference of squares:  $x^2 - 16 = (x - 4)(x + 4)$ .
- 10 Factor  $2x^2 + 8x$  completely.  
Factor out the greatest common factor:  $2x^2 + 8x = 2x(x + 4)$ .

- 11 Factor  $x^3 - 4x$  completely.

Factor out  $x$ :  $x^3 - 4x = x(x^2 - 4)$ , and then factor the remaining difference of squares:  $x(x - 2)(x + 2)$ .

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## Polynomial division and the remainder theorem

- 12 Find the remainder when  $f(x) = x^3 - 2x^2 + x - 5$  is divided by  $(x - 2)$ , using the remainder theorem.

By the remainder theorem, the remainder equals  $f(2)$ :  $f(2) = 8 - 8 + 2 - 5 = -3$ .

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## Exponential growth vs. polynomial growth

- 13 Two functions are given:  $f(x) = x^3$  and  $g(x) = 2^x$ . Explain which function eventually grows faster as  $x$  increases, and estimate the crossover point using the values  $f(10) = 1000$ ,  $g(10) = 1024$ ,  $f(8) = 512$ ,  $g(8) = 256$ .

Exponential functions eventually always outgrow polynomial functions as  $x \rightarrow \infty$ , regardless of the polynomial's degree, because the exponential's rate of increase itself keeps growing while the polynomial's relative rate of increase slows. Comparing the given values: at  $x = 8$ ,  $f(8) = 512 > g(8) = 256$  (polynomial ahead), but by  $x = 10$ ,  $g(10) = 1024 > f(10) = 1000$  (exponential has overtaken it) — so the crossover happens between  $x = 8$  and  $x = 10$ .

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## Modeling with exponential functions

- 14 An investment of 5000 dollars grows according to  $A(t) = 5000(1.06)^t$ , where  $t$  is the number of years. Find the value of the investment after 5 years, to the nearest dollar, and state the annual growth rate as a percentage.

$A(5) = 5000(1.06)^5 = 5000(1.3382) \approx 6691$  dollars. The annual growth rate is 6%, since the growth factor is  $1.06 = 1 + 0.06$ .

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## Applying the zero product property to polynomial equations

- 15 Solve  $x^3 - 5x^2 + 6x = 0$  by factoring completely and applying the zero product property.

Factor out  $x$ :  $x(x^2 - 5x + 6) = 0$ , then factor the quadratic:  $x(x - 2)(x - 3) = 0$ . By the zero product property,  $x = 0$ ,  $x = 2$ , or  $x = 3$ .

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## Multiplying binomials and trinomials

- 16 Expand  $(x + 2)(x^2 - 3x + 4)$ .

Distribute each term:

$$x(x^2 - 3x + 4) + 2(x^2 - 3x + 4) = x^3 - 3x^2 + 4x + 2x^2 - 6x + 8 = x^3 - x^2 - 2x + 8.$$

- 17 Expand  $(2x - 3)(x + 5)$ .  
 $(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$ .
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### Comparing exponential models

- 18 Two savings accounts grow according to  $A(t) = 1000(1.03)^t$  and  $B(t) = 1000(1.05)^t$ . Explain which account grows faster, and find the difference in value after 10 years, to the nearest dollar.
- Account  $B$  grows faster because it has a larger growth factor ( $1.05 > 1.03$ , meaning a higher annual interest rate). After 10 years:  $A(10) = 1000(1.03)^{10} \approx 1344$ ,  $B(10) = 1000(1.05)^{10} \approx 1629$ .  
Difference:  $1629 - 1344 = 285$  dollars.
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### Simplifying expressions with multiple exponent rules

- 19 Simplify  $\frac{(x^3y^2)^2}{x^4y}$ .  
Expand the numerator:  $(x^3y^2)^2 = x^6y^4$ . Divide:  $\frac{x^6y^4}{x^4y} = x^{6-4}y^{4-1} = x^2y^3$ .
- 20 Simplify  $\left(\frac{2x^2}{y}\right)^3$ .  
 $\left(\frac{2x^2}{y}\right)^3 = \frac{2^3(x^2)^3}{y^3} = \frac{8x^6}{y^3}$ .
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### Half-life and decay applications

- 21 A radioactive substance has a half-life of 8 years, starting with 200 grams. Write a model for the remaining amount after  $t$  years, and find the amount remaining after 24 years.
- Model:  $A(t) = 200\left(\frac{1}{2}\right)^{t/8}$ . After 24 years:  $A(24) = 200\left(\frac{1}{2}\right)^{24/8} = 200\left(\frac{1}{2}\right)^3 = 200\left(\frac{1}{8}\right) = 25$  grams.